

Waves and Rotations

Sine, cosine, and turning arrows · video 1:48

This lesson has one big idea: **sine and cosine come from a point moving around a circle**. Its height traces a wave that repeats every full turn, 2π , and the same two functions can turn an arrow by any angle without changing its length. Episode 4 uses both: waves of different frequencies to mark positions, and rotations that turn relative position into a relative angle.

How to use this guide: master one concept before moving to the next.

1. Watch the whole video once, without stopping.
2. Then take the concepts **in order**. For each one: read the short explanation, look at the picture from the video (▶ is the time in the video), and answer every question in its *Check* without looking anything up.
3. Check your answers in the **Answers** section at the end (page 9).
4. **Pass mark: every question in the check correct.** If you missed one, rewatch that concept's video segment, reread the explanation, and redo only the questions you missed. Tick the concept off below, then move on.

It is strongly suggested that you **master each concept before starting the next one**. Each concept builds on the previous ones.

Concepts in this lesson

CONCEPT	VIDEO SEGMENT	QUESTIONS	MASTERED
1. Sine and cosine on the unit circle	▶ 0:08-0:25	4	☐
2. The sine wave, and a full turn of 2π	▶ 0:25-0:46	4	☐
3. Frequency: how fast it repeats	▶ 0:47-0:57	4	☐
4. Many waves: a fingerprint for every position	▶ 0:58-1:06	4	☐
5. Rotating an arrow	▶ 1:06-1:18	4	☐
6. Relative angles, and rotation in code	▶ 1:18-1:37	3	☐

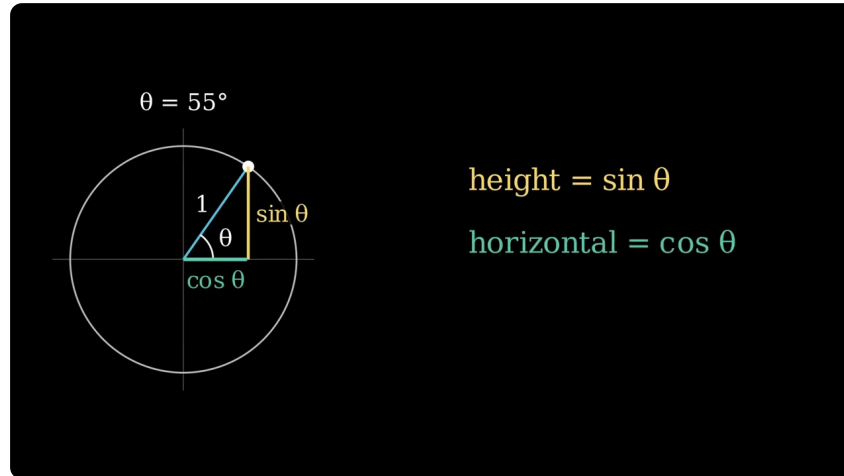
Before you start: concepts 5 and 6 use 2-D vectors and their length from F01 (Vectors) and matrix-times-vector from F03 (Matrices). The code uses NumPy; if it is new to you, F11 (NumPy in Three Minutes) helps. These ideas are used in episode 4 (Where Am I? Position): sine waves mark positions, and rotations appear as RoPE. Set your calculator to **radians** for the Checks, unless a question uses degrees ($^\circ$).

CONCEPT 1 OF 6

Sine and cosine on the unit circle

▶ Video 0:08-0:25

Picture a point moving around a circle of **radius 1**. At any angle θ , its **height** is the **sine** of the angle, $\sin \theta$, and its **horizontal position** is the **cosine**, $\cos \theta$. So the point sits at **(cos θ , sin θ)**.



▶ 0:24 A point on a circle of radius 1: its height is $\sin \theta$, its horizontal position is $\cos \theta$.

Some angles to know: at 0° the point is at (1, 0), at 90° at the top, (0, 1), and at 30° at (0.866, 0.5). Because the radius is 1, Pythagoras (F01) gives $\cos^2 \theta + \sin^2 \theta = 1$ at every angle: for 30° , $0.75 + 0.25 = 1$.

KEY IDEA

On a circle of radius 1, the point at angle θ is **(cos θ , sin θ)**: cosine is the horizontal position, sine is the height. Both always stay between -1 and 1 .

Check 1: Sine and cosine on the unit circle

Answer all 4 questions before looking at the answers. Pass mark: all correct.

1.1 MULTIPLE CHOICE

The point is at the very top of the circle, at 90° . What are $\cos 90^\circ$ and $\sin 90^\circ$?

- A. $\cos = 1$, $\sin = 0$
- B. $\cos = 0$, $\sin = 1$
- C. $\cos = 1$, $\sin = 1$
- D. $\cos = 0.5$, $\sin = 0.5$

1.2 CALCULATION

Where is the point at 180° ? Give (cos 180° , sin 180°).

1.3 TRUE OR FALSE

"For some angle θ , $\sin \theta = 1.5$."

True ☐ False ☐

1.4 CALCULATION

At 60° the height is $\sin 60^\circ \approx 0.866$. Use $\cos^2 \theta + \sin^2 \theta = 1$ to find $\cos 60^\circ$.

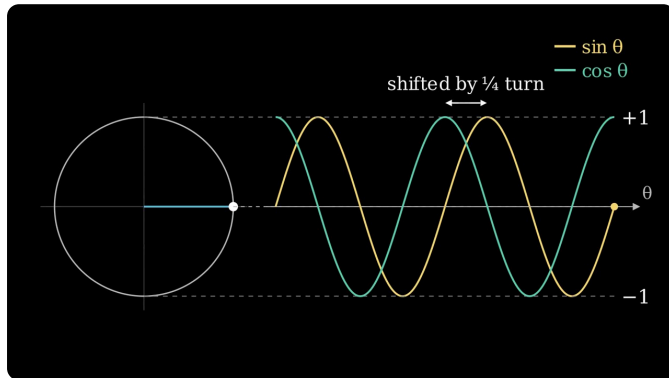
CONCEPT 2 OF 6

The sine wave, and a full turn of 2π

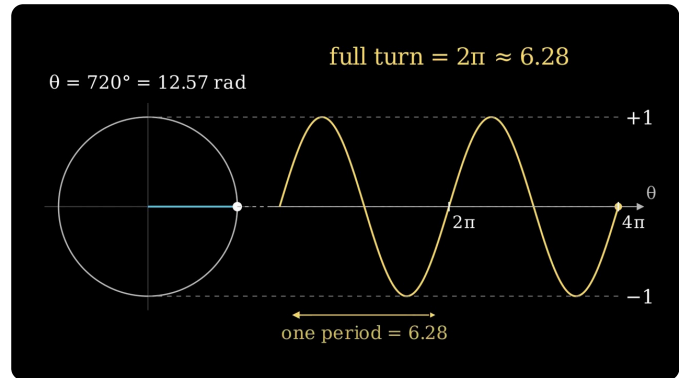
▶ Video 0:25–0:46

Let the angle keep growing and plot the height as you go: that is the **sine wave**. It goes up to 1, down to -1 , and repeats forever, because after a full turn the point is back where it started. The **cosine** is the same wave, shifted by a quarter turn.

Angles are usually measured in **radians**: a full turn (360°) is $2\pi \approx 6.28$, so the wave repeats every 6.28 units, one **period**. Half a turn (180°) is $\pi \approx 3.14$; a quarter turn (90°) is $\pi/2 \approx 1.57$.



▶ 0:36 Plot the height: the sine wave. Cosine is the same wave, $1/4$ turn shifted.



▶ 0:46 A full turn is $2\pi \approx 6.28$, so the wave repeats every 6.28.

KEY IDEA

The sine wave is the moving point's height, plotted against the angle. It repeats every full turn: $360^\circ = 2\pi \approx 6.28$ radians.

Check 2: The sine wave, and a full turn of 2π

Answer all 4 questions before looking at the answers. Pass mark: all correct.

2.1 CALCULATION

Convert to radians: (a) 180° (b) 90° (c) 30° .

2.2 TRUE OR FALSE

" $\sin(1)$ and $\sin(1 + 2\pi)$ are equal."

True ☐ False ☐

2.3 MULTIPLE CHOICE

The cosine wave is...

- A. the sine wave flipped upside down
- B. the sine wave shifted by a quarter turn
- C. the sine wave, twice as fast
- D. a straight line

2.4 CALCULATION

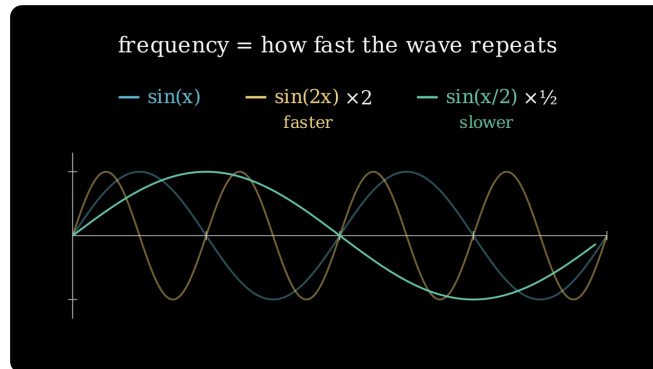
How many full periods does $\sin \theta$ complete between $\theta = 0$ and $\theta = 4\pi \approx 12.57$?

CONCEPT 3 OF 6

Frequency: how fast it repeats

▶ Video 0:47–0:57

Frequency is how fast a wave repeats. Multiply the input by a bigger number and the wave oscillates faster: $\sin(2x)$ repeats twice as often as $\sin(x)$, every $\pi \approx 3.14$ instead of every 6.28 . Multiply by a smaller number and it slows down: $\sin(x/2)$ repeats only every $4\pi \approx 12.57$. The height still goes from -1 to 1 .



▶ 0:57 $\times 2$: twice as fast. $\times \frac{1}{2}$: half as fast.

KEY IDEA

$\sin(w \cdot x)$ repeats every $2\pi \div w$. A bigger w means a faster wave (a higher frequency); a smaller w , a slower one.

Check 3: Frequency: how fast it repeats

Answer all 4 questions before looking at the answers. Pass mark: all correct.

3.1 CALCULATION

How long is one period of $\sin(3x)$?

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3.2 PUT IN ORDER

Order these waves by the length of their period, shortest first: $\sin(x/2) \cdot \sin(4x) \cdot \sin(x) \cdot \sin(2x)$.

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3.3 TRUE OR FALSE

“Multiplying the input by 2 makes the wave twice as tall.”

True ☐ False ☐

3.4 CALCULATION

You want a wave that repeats exactly every 100 units. What should w be in $\sin(w \cdot x)$?

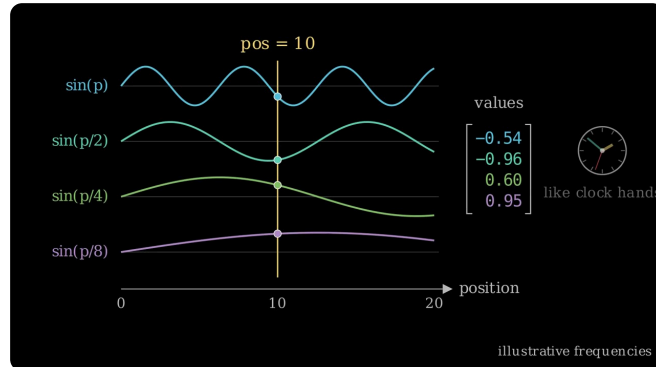
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CONCEPT 4 OF 6

Many waves: a fingerprint for every position

▶ Video 0:58–1:06

Combine several waves with different frequencies, here $\sin(p)$, $\sin(p/2)$, $\sin(p/4)$ and $\sin(p/8)$, and read them all at the same position p . Every position gets its **own combination of values**: at $p = 3$ it is 0.14, 1.00, 0.68, 0.37; at $p = 10$ it is -0.54 , -0.96 , 0.60, 0.95.



▶ 1:05 At position 10 the four waves read -0.54 , -0.96 , 0.60, 0.95.

It works like the hands of a clock. The fast wave changes a lot from one position to the next but soon repeats; the slow waves change little but take long to repeat. Together they tell positions apart.

KEY IDEA

One wave repeats, so on its own it cannot tell far-apart positions apart. Several waves with different frequencies give every position its own list of values.

Check 4: Many waves: a fingerprint for every position

Answer all 4 questions before looking at the answers. Pass mark: all correct.

4.1 CALCULATION

With a calculator in radian mode, compute $\sin(p/4)$ at $p = 10$. Which of the four values in the picture is it?

4.2 MULTIPLE CHOICE

Why not mark positions with the fast wave $\sin(p)$ alone?

- A. It repeats every 6.28 positions, so far-apart positions can get almost the same value
- B. It is always 0 at whole-number positions
- C. It changes too slowly to tell neighbors apart
- D. Sine cannot take whole numbers as input

4.3 CALCULATION

How many positions does the slowest wave, $\sin(p/8)$, take to repeat?

4.4 TRUE OR FALSE

“From one position to the next, $\sin(p)$ changes more than $\sin(p/8)$ does.”

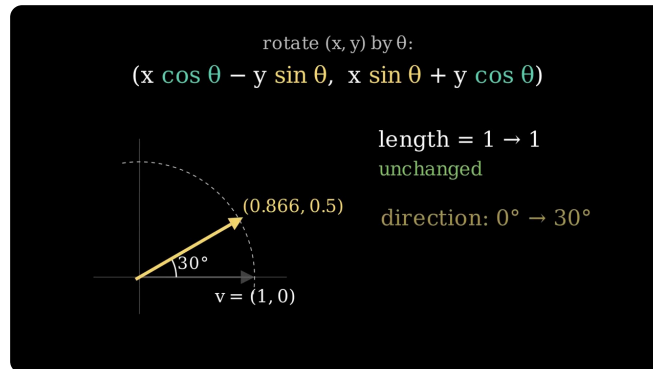
True ☐ False ☐

CONCEPT 5 OF 6

Rotating an arrow

▶ Video 1:06–1:18

Sine and cosine also **rotate** arrows. To turn a 2-D vector (x, y) by an angle θ , mix its two numbers with $\cos \theta$ and $\sin \theta$: **$(x \cos \theta - y \sin \theta, x \sin \theta + y \cos \theta)$** . The **length never changes**, only the direction. The video turns $v = (1, 0)$ by 30° : $(1 \times 0.866 - 0 \times 0.5, 1 \times 0.5 + 0 \times 0.866) = \mathbf{(0.866, 0.5)}$. Its length is still $\sqrt{(0.75 + 0.25)} = 1$; only the direction went from 0° to 30° .



▶ 1:17 $(1, 0)$ turned by 30° : $(0.866, 0.5)$, length still 1.

KEY IDEA

Rotate (x, y) by θ : **$(x \cos \theta - y \sin \theta, x \sin \theta + y \cos \theta)$** . The direction turns by θ ; the length stays the same.

Check 5: Rotating an arrow

Answer all 4 questions before looking at the answers. Pass mark: all correct.

5.1 CALCULATION

Rotate $(0, 1)$ by 30° . ($\cos 30^\circ \approx 0.866$, $\sin 30^\circ = 0.5$.)

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5.2 CALCULATION

Rotate $(3, 4)$ by 90° ($\cos 90^\circ = 0$, $\sin 90^\circ = 1$). Check that the length is unchanged.

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.....

5.3 TRUE OR FALSE

"Rotating a vector by a large enough angle makes it longer."

True ☐ False ☐

5.4 MULTIPLE CHOICE

Rotating $(2, 0)$ by 30° gives...

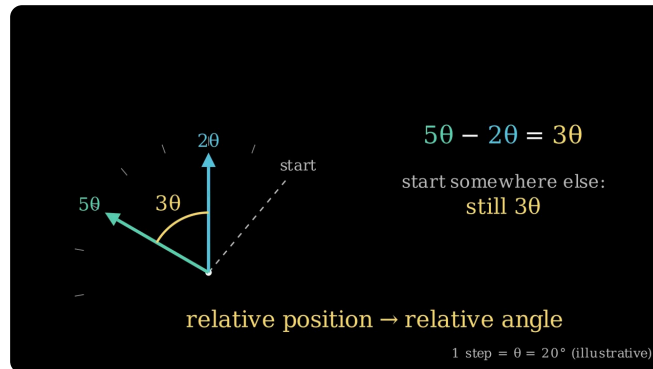
- A. $(1.732, 1)$
- B. $(0.866, 0.5)$
- C. $(2, 0.5)$
- D. $(1, 1.732)$

CONCEPT 6 OF 6

Relative angles, and rotation in code

▶ Video 1:18-1:37

Rotate one arrow by two steps (2θ) and another by five (5θ). The angle between them is $5\theta - 2\theta = 3\theta$, no matter where both started. So if each position turns an arrow by its own number of steps, **relative position becomes a relative angle**: positions 5 and 2 end up 3 steps apart, and so do 12 and 9.



▶ 1:28 Turn by 2θ and by 5θ : they are 3θ apart, from any start.

In code, a rotation is a 2×2 matrix built from the cosine and the sine. Multiplying it with a vector (F03) computes exactly the formula of concept 5, and the length stays exactly 1.

```
theta = np.pi / 6          # 30 degrees
R = np.array([[np.cos(theta), -np.sin(theta)],
              [np.sin(theta),  np.cos(theta)]])
v = np.array([1.0, 0.0])
R @ v                       # array([0.866, 0.5])
np.linalg.norm(R @ v)      # 1.0
```

KEY IDEA

Turning by 2θ and by 5θ leaves two arrows **3θ apart**, from any starting direction: only the difference in steps shows. A rotation is a 2×2 matrix of cos and sin.

Check 6: Relative angles, and rotation in code

Answer all 3 questions before looking at the answers. Pass mark: all correct.

6.1 CALCULATION

Use the video's step, $\theta = 20^\circ$. (a) At which angles do the 2θ and 5θ arrows point? (b) What is the angle between them? (c) Now start both 50° further round. Answer (a) and (b) again.

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6.2 MULTIPLE CHOICE

Which pair of positions gives the same relative angle as positions 5 and 2?

- A. 7 and 1
- B. 10 and 7
- C. 5 and 3
- D. 6 and 2

6.3 HANDS-ON CODE

Try it yourself. The code wraps the video's matrix in a function `rot`. Predict, then run: (a) `rot(theta) @ np.array([0.0, 1.0])`. (b) `rot(theta) @ rot(theta) @ v`: by what angle has `v` turned? (c) `np.linalg.norm(rot(theta) @ np.array([3.0, 4.0]))`.

```
import numpy as np
np.set_printoptions(precision=3, suppress=True)

def rot(theta):
    return np.array([[np.cos(theta), -np.sin(theta)],
                     [np.sin(theta),  np.cos(theta)]])

theta = np.pi / 6                # 30 degrees
v = np.array([1.0, 0.0])
print(rot(theta) @ v)             # [0.866 0.5  ]
```

ANSWERS

Answers and explanations

Mark each answer right or wrong. If a concept has any wrong answer, rewatch its segment (▶) and redo the questions you missed before moving on.

1. Sine and cosine on the unit circle · rewatch ▶ 0:08-0:25

1.1 B. At the top the point is (0, 1): no horizontal offset, full height.

1.2 (-1, 0). Half a turn puts the point on the far left, at height 0.

1.3 False. $\sin \theta$ is the height of a point on a circle of radius 1, so it is always between -1 and 1.

1.4 0.5. $\cos^2 60^\circ = 1 - 0.75 = 0.25$, and $\sqrt{0.25} = 0.5$ (positive, because at 60° the point is right of the center).

2. The sine wave, and a full turn of 2π · rewatch ▶ 0:25-0:46

2.1 (a) $\pi \approx 3.14$ (b) $\pi/2 \approx 1.57$ (c) $\pi/6 \approx 0.52$. $360^\circ = 2\pi$, so $180^\circ = \pi$ and $90^\circ = \pi/2$. 30° is a twelfth of a turn: $2\pi \div 12 = \pi/6$ (the video's code writes `np.pi / 6`).

2.2 True. Adding a full turn brings the point back to the same place (both ≈ 0.841).

2.3 B. It has the same shape, shifted by a quarter turn ($\pi/2$).

2.4 2. One period is 2π , and $4\pi \div 2\pi = 2$. The video draws exactly these two.

3. Frequency: how fast it repeats · rewatch ▶ 0:47-0:57

3.1 $2\pi/3 \approx 2.09$. Three times as fast, so a third of 6.28.

3.2 $\sin(4x) \rightarrow \sin(2x) \rightarrow \sin(x) \rightarrow \sin(x/2)$. Periods 1.57, 3.14, 6.28 and 12.57: the bigger the multiplier, the shorter the period.

3.3 False. It makes the wave repeat twice as fast. The height still only goes from -1 to 1.

3.4 $w = 2\pi \div 100 \approx 0.0628$. The period is $2\pi \div w$, so $w = 2\pi \div \text{period}$.

4. Many waves: a fingerprint for every position · rewatch ▶ 0:58-1:06

4.1 $\sin(2.5) \approx 0.60$: the third value. $10 \div 4 = 2.5$, and $\sin(2.5) \approx 0.598$.

4.2 A. For example $\sin(0) = 0$ and $\sin(44) \approx 0.02$, but the slower $\sin(p/4)$ tells them apart: 0 vs -1.00.

4.3 About 50.3. $2\pi \div (1/8) = 16\pi \approx 50.3$.

4.4 True. $\sin(p)$ is the fast wave, like the second hand; $\sin(p/8)$ moves slowly, like the hour hand.

5. Rotating an arrow · rewatch ▶ 1:06-1:18

5.1 (-0.5, 0.866). $x = 0$, $y = 1$: $(0 - 1 \times 0.5, 0 + 1 \times 0.866)$. The arrow pointed at 90° and now points at 120° .

5.2 (-4, 3); length 5 before and after. $(3 \times 0 - 4 \times 1, 3 \times 1 + 4 \times 0) = (-4, 3)$. $\sqrt{(9 + 16)} = \sqrt{(16 + 9)} = 5$.

5.3 False. A rotation only changes the direction, never the length.

5.4 A. Twice the video's result, $2 \times (0.866, 0.5)$: the length stays 2. D is a turn of 60° .

6. Relative angles, and rotation in code · rewatch ► 1:18-1:37

6.1 (a) 40° and 100° . (b) 60° . (c) 90° and 150° ; still 60° . The shared start adds the same 50° to both, so the difference $3\theta = 60^\circ$ does not change.

6.2 B. $10 - 7 = 3$ steps, like $5 - 2$. The others are 6, 2 and 4 steps apart.

6.3 (a) $[-0.5 \ 0.866]$ · (b) $[0.5 \ 0.866]$, 60° · (c) 5.0 (a) The same as question 5.1: $(0, 1)$ turned from 90° to 120° . (b) Two turns of 30° make one turn of 60° , and $(\cos 60^\circ, \sin 60^\circ) = (0.5, 0.866)$: rotations add up, which is exactly why $5\theta - 2\theta = 3\theta$ works. (c) $(3, 4)$ has length 5, and rotating never changes the length.